UNIVERSITY of TASMANIA

AMC

Australian Maritime College

National Centre for Maritime Engineering & Hydrodynamics

Australian Maritime College

JEE523/JEE529
DESIGN OF MARINE MACHINERY SYSTEMS

Semester 1, 2022

Unit Outline

Unit Coordinator: Associate Professor Norman Lawrence

CRICOS Provider Code: 00586B

CONTACT DETAILS

Unit coordinator & lecturer

Unit coordinator:	Norman Lawrence
Campus:	Newnham
Email:	N.Lawrence@utas.edu.au
Phone:	0400 132 237
Room location and number:	Swanson Building, G68
Consultation hours:	By email or 4-5 pm Thursdays, G68

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COVID Safety

To ensure the safety and wellbeing of our University and the broader community, it is important that we follow the guidelines for coming onto campus or participating in UTAS activities or events.

Please consult the Student Coronavirus updates page for up-to-date information on COVID-19 and anything you need to do as a UTAS student.

WHAT IS THE UNIT ABOUT?

Unit description

This unit equips students with an understanding of the functional purpose of the machinery systems used on board a variety of ships. This knowledge can then be applied to obtaining effective engineering design solutions for these systems, so that they can meet their functional purposes within national and international statutory and regulatory frameworks.

Intended Learning Outcomes

As per the Assessment and Results Policy 1.3, your results will reflect your achievement against specified learning outcomes.

On completion of this unit, you will be able to:

1. Explain the design features and practical constraints of marine machinery systems and components.
2. Analyse fluid and air flows in duct and pipe systems to determine appropriate pump/fan and cooler duties.
3. Select and specify the appropriate equipment for marine power plant with due regard to reliability and maintainability.
4. Design ship's machinery, piping, and systems, with due consideration to classification and statutory regulations.
5. Apply considerations of occupational health and safety and environmental protection issues to the design of marine machinery systems.

Graduate Statement

Successful completion of this unit supports your development of course learning outcomes, which describe what a graduate of a course knows, understands and is able to do. Course learning outcomes published in the Course Rules. This document is available on the NCMEH Course Information website. Course learning outcomes are developed with reference to national discipline standards, Australian Qualifications Framework (AQF), any professional accreditation requirements and the University of Tasmania's Graduate Statement.

> The University of Tasmania experience unlocks the potential of individuals. Our graduates are equipped and inspired to shape and respond to the opportunities and challenges of the future as accomplished communicators, highly regarded professionals and culturally competent citizens in local, national, and global society. University of Tasmania graduates acquire subject and multidisciplinary knowledge and skills, and develop critical and creative literacies and numeracies and skills of inquiry. They demonstrate the ability to apply this knowledge in changing circumstances. Our graduates recognise and critically evaluate issues of social responsibility, ethical conduct and sustainability, are entrepreneurial and creative, and are mindful of their own wellbeing and that of the community. Through respect for diversity and by working in collaborative ways, our graduates reflect the values of the University of Tasmania.

Alterations to the unit as a result of student feedback

Student feedback for this unit has been very positive and the unit required no alterations resulting from student feedback. As part of ongoing unit development, several case studies have been created to help consolidate student understanding of air flow, air conditioning and reliability centred maintenance.

Prior knowledge &/or skills

General undergraduate engineering knowledge, especially Heat Transfer, Thermodynamics and Fluid mechanics.

HOW WILL I BE ASSESSED?

Assessment schedule

Assessment task	Date due	Percent weighting	Links to Intended Learning Outcomes
Class Test	10th April 2022	12.5%	1,2
Discussions	Various	12.5%	1,3,4,5
Virtual Conference Paper	30th April 2022	25%	3,4,5
Design Report	29th May 2022	50%	1,2,3,4

Assessment details

Class Test

Task description	An open book class test is scheduled as shown in the Assessment and Unit Schedule and is worth 12.5% of the total unit marks. This will cover theory and applications covered up to and including week 5 material. The class test will be available electronically at 9.00 am on Friday 8th April and individual answer scripts will be submitted using the prescribed format electronically on or before 11.00 pm Sunday 10th April.	
	Criterion	Measures Intended Learning Outcome:
Criterion 1	Ability to read a question and extract the relevant information and formulae	1
Criterion 2	Communicate thought processes and ability to develop a solution strategy	2
Criterion 3	Demonstrate engineering and mathematical analytical skills.	2
Task length	3 hours	

Discussions

Task description	Online discussions form an integral part of the program in this unit. Online participation by all members helps to create a sense of learning community and online discussions give us a window on collective and individual thinking. Expressing yourself in succinct words helps clarify your own thinking and assists in testing that thinking with fellow students and staff. In short, discussions are a great facilitator of critical thinking, a hallmark of postgraduate study	
	Criterion	Measures Intended Learning Outcome:
Criterion 1	Number and frequency of postings, Quality of postings	1,5
Criterion 2	Value of individual contributions	3,4
Criterion 3	Analysis and critical thinking,	3,4
Criterion 4	Integration, Resolution, Contribution towards group functioning and outcomes.	1,5
Task length	2 hours per topic	

Virtual Conference Paper

Task description	The virtual conference requires each student to research an agreed topic relevant to ship machinery systems and submit (30th April) their individual assignment using the prescribed format electronically and the assessed work will be returned electronically. Students will participate in a one week virtual conference (2rd-8th May), posing questions to authors of other papers and in turn answering questions from other authors regarding their own paper.	
	Criterion	Measures Intended Learning Outcome:
Criterion 1	Quality of research	Criterion 1
Criterion 2	Quality of analysis and evaluation of the issue chosen	Criterion 2
Criterion 3	Logical structure and flow of ideas; overall clarity of writing and conformance to appropriate referencing standards and paper format requirements	Criterion 3
Criterion 4	Quality of participation and leadership in online discussion	Criterion 4

Design Report

Task description	The design study is an individual design project and the report will be submitted using the prescribed format electronically.	
	Criterion	Measures Intended Learning Outcome:
Criterion 1	General description	1
Criterion 2	Reading, Knowledge of topic	4
Criterion 3	Articulation of argument	3,4
Criterion 4	Analytical and evaluative skills	2,3
Criterion 5	Expression and presentation appropriate to the discipline	1

How your final result is determined

To pass this unit you need to achieve an overall mark of 50%.

Your grade will be determined in the following way:

Grade	Fail	Pass	Credit	Distinction	High Distinction
Mark (%)	<50	50-59	60-69	70-79	80+

Submission of assignments

All coursework must have the AMC-NCMEH Assignment Cover Sheet attached which can be downloaded from the NCMEH Course Information website.

All assignments and reports must be typed and completed using Word, Excel, approved Engineering drawing software and include the relevant theory, illustrations, results, analysis, and conclusion.

Assignments and reports must be submitted electronically using the units MyLO dropbox. The assessed work will be returned electronically.

Please remember that you are responsible for lodging your coursework on or before the due date. It is strongly recommend that you keep a copy; even in the most 'perfect' of systems, items sometimes go astray.

Requests for extensions

Extensions will only be granted on medical or compassionate grounds. Requests for extensions should be made via email to the lecturer prior to the due date. Medical certificates or other evidence must be included (electronically or the hard copy mailed) and must contain information which justifies the extension sought.

Penalties

Late assignments which have not been granted an extension will, at the lecturer's discretion, be penalised by deducting 5 per cent of total marks for each calendar day overdue. Late submissions will not be accepted more than 10 calendar days after the due date, or after assignments have been returned to other students, whichever is shorter.

Failure to adhere with the WH&S standards whilst taking part in any assessed activity that involves field trips and/or that requires the use of UTAS or AMC facilities will result in the following penalties:

- ***first offense:*** *5% penalty (applicable to the whole team in team projects);*
- ***second offense by student (or another member of the same team in team projects):*** *20% penalty (applicable to the whole team in team projects); and*
- ***third offense:*** *a fail grade in the assessment.*

Review and appeal of academic decisions

For processes guiding the review and appeal of University academic decisions (related to credit, assessment, final unit grades, progress status and special consideration) see the *Review and Appeal of Academic Decisions Procedure*.

If you have questions about, or problems with, your assessment you should, in the first instance, discuss this with the following people:

(1) Unit Coordinator.

(2) Course Coordinator.

(3) Director, NCMEH.

Academic integrity

What is academic integrity?

At the University of Tasmania, academic integrity requires all students to act responsibly, honestly, ethically, and collegially when using, producing, and communicating information with other students and staff members. The University community is committed to upholding the Statement on Academic Integrity.

Breaches of academic integrity such as plagiarism, contract cheating, collusion and so on are counter to the fundamental values of the University. A breach is defined as being when a student:

a) fails to meet the expectations of academic integrity; or
b) seeks to gain, for themselves or for any other person, any academic advantage or advancement to which they or that other person is not entitled; or
c) improperly disadvantages any other member of the University community.

The University and any persons authorised by the University may submit your assessable works to a text matching service, to obtain a report on possible breaches such as plagiarism

or contract cheating. Substantiated breaches can result in a range of sanctions which are outlined in the Student Academic Integrity Ordinance.

More information is available from the Academic Integrity site for students on the Student Portal.

Academic Integrity Training Module

As part of the University's educative approach to academic integrity, there is a short Academic Integrity Training Module on MyLO that all students are required to complete.

Completion of the module allows you to demonstrate your understanding of what constitutes a breach of academic integrity.

All commencing students (pre-degree through to higher degree by research) are required to complete the Academic Integrity module available through MyLO. If you do not complete this module your final unit results will be withheld. You should aim to complete the module within the first few weeks of commencing study at the University.

Academic referencing

In your written work you will need to support your ideas by referring to scholarly literature, works of art and/or inventions. It is important that you understand how to correctly refer to the work of others and maintain academic integrity.

Failure to appropriately acknowledge the ideas of others may constitute a breach of academic integrity.

The appropriate referencing style for this unit is Harvard author-date

The University library provides information on presentation of assignments, including referencing styles and should be referred to when completing tasks in this unit.

For further information, see the Academic Integrity site for students on the Student Portal.

WHAT LEARNING OPPORTUNITIES ARE THERE?

MyLO

MyLO is the online learning environment at the University of Tasmania. All teaching necessary for the successful completion of this unit is contained within MyLO unit for JEE523/JEE529.

Getting help with MyLO

It is important that you are able to access and use MyLO as part of your study in this unit. To find out more about the features and functions of MyLO, and to practice using them, visit the Getting Started in MyLO unit.

For access to information about MyLO and a range of step-by-step guides in pdf, word and video format, visit the MyLO Student Support page on the University website.

If something is not working as it should, contact the Service Desk (phone 6226 2600 or request ITS help online through the Service Portal).

Resources

Required readings

None

Recommended reading

Australian Standards, 1921-1976, Air conditioning and ventilation in ships' cabins and living spaces of merchant ships.

Burcher R. and Rydill, L., *Concepts in Submarine Design*, 2nd Ed., Cambridge University Press, 1998.

DEF(AUST), 5000, Volume 4, Part 3, Heating Ventilation and Air Conditioning.

Douglas, J.F., Gasiorek, J.M. and Swaffield, J.A., *Fluid Mechanics*, 4th Ed., Prentice Hall, 2001.

Gabler, U, *Submarine Design*, 2nd Ed., Bernard and Graefe Verlag, 2000.

ISO 7547:2002 ***Ships and marine technology - Air-conditioning and ventilation of accommodation spaces - Design conditions and basis of calculations*.**

Hodge, B.K. and Taylor, R.P., *Analysis and Design of Energy Systems,* 3rd Ed. Prentice Hall, New Jersey, 1999.

Janna, W.S., *Design of Fluid Thermal Systems*, 2nd Ed, PWS Publishing Company, Boston, MA, 1998.

Meier-Peter, H. and Bernhardt, F., *Compendium Marine Engineering,* Seehafen Verlag, Hamburg, 2009

Molland, A.F., ***The Maritime Engineering Reference Book***, Butterworth-Heinemann, Oxford, 2008.

McGeorge, H.D., *Marine Auxiliary Machinery*, 7th Ed, Butterworths-Heinemann, Oxford, 2004 (latest reprint). Available as an ebook from the university library.)

McQuiston, F.C., Parker, J. D. and Spitler, J. D. 2000 *Heating, Ventilating and Air Conditioning; Analysis and Design,* 5ed, Wiley, New York.
Harrington R.L., *Marine Engineering*, SNAME, N.J. 1992.

Taylor, D.A., *Introduction to Marine Engineering*, 2nd Ed Butterworths-Heinemann, Oxford, 2002.

Rules and Regulations of Classification Societies for the Marine Industry.

Woud, H.K. and Stapersma, D., *Design of Propulsion and Electric Power Generation Systems*, IMarEST Publications, London, 2003

Reading Lists

You can access the Reading List for this unit from the link in MyLO, or by going to the Reading Lists page on the University Library website.

Equipment, materials, software, accounts

Materials to be provided by the student

- A computer with internet access, Microsoft Office Suite and Adobe Acrobat reader. Access to a CAD program would be useful (but not mandatory) for the Design Report.

Activities

Learning expectations

The University is committed to high standards of professional conduct in all activities, and holds its commitment and responsibilities to its students as being of paramount importance. Likewise, it holds expectations about the responsibilities students have as they pursue their studies within the special environment the University offers.

Students are expected to participate actively and positively in the teaching/learning environment. They must attend classes when and as required, strive to maintain steady progress within the subject or unit framework, comply with workload expectations, and submit required work on time.

Details of teaching arrangements

This unit has both on and off campus students. Off campus study does not involve any formal lectures or tutorials. On campus students have a workshop and tutorial on Thursday afternoons. It is suggested that every student works through the modules according to a definite schedule, aiming to complete one module per week. All learning and assessment material for the unit is to be found on the MyLO website.

The unit schedule that follows is a suggested timetable for both on and off campus students, as it follows the standard AMC/UTas teaching semester.

CLASS	DAY	TIME	LOCATION	GROUP
Weekly module	Thursday		online	All
Workshop	Thursday	2.00 pm	G81	On campus
Tutorial	Thursday	3.00 pm	G81	On campus

Topics covered

Design Methodology and Regulations

Statutory and classification regulations for marine machinery and systems. Occupational health and safety, efficiency and environmental considerations for marine machinery and associated systems. Optimisation of machinery space lay out. Reliability, maintainability and systems engineering methodology.

Marine Auxiliary Systems

Electric generators, engine and auxiliary space requirements and general arrangements. Ballast and bilge systems, fire and general service systems, fresh water and sea water cooling systems, centralized cooling system, waste management, lubricating oil and fuel oil systems, compressed air systems, oil separators, air compressors and pumps. Hydraulic systems.

Marine Electrical Systems

The essential features of a ship's electrical system. Three phase generators, switchboards and general switching arrangements. Automatic voltage regulation, synchronising and load sharing. Modern power management systems and electric load profiling. Electric machinery. Three phase induction motor and synchronous motor, construction, principles and operational features. Speed control and starting arrangements including programmable logic controllers. Special equipment for hazardous areas.

Fluid Thermal Systems

Pump theory and system losses, cavitation, pump-pipe system characteristics and operating point. Heat exchanger theory, practice and specification, including shell & tube and plate heat exchangers. Heat exchanger thermal design using both LMTD and NTU-effectiveness methods. Air conditioning systems, comfort and health, moist air and conditioning process. Design principles for ventilation, refrigeration, chilled water and heating systems. Heating and cooling loads, energy calculations, space air distribution including ducts and trunking

Teaching and learning strategies

All learning and assessment material for the unit is to be found on the MyLO website. The unit has both on and off campus students. It is suggested that every student works through the modules aiming to complete one module per week.

The unit has been carefully designed to create the sense of a learning community through online participation by all students. This is especially important when class members only meet face-to-face infrequently, or not at all.

- The open book class test consolidates the foundational knowledge of the early modules.
- Online discussions present an opportunity to generally share ideas and explore practical and topical applications of theory presented in the learning modules.
- The virtual conference provides a window on collective and individual thinking through the research of topic relevant to the course and presenting it online for discussion.

The design study allows students to apply the core teaching material through an industry relevant task.

Work Health and Safety (WHS)

The University is committed to providing a safe and secure teaching and learning environment. In addition to specific requirements of this unit you should refer to the University's Safety and Wellbeing webpage and policy.

Communication

News and announcements may be posted to MyLO News, and students will be expected to be aware of the content of such posts within 48 hours of them being posted.

Students are also expected to check their UTAS email very regularly (once a day) for important announcements.

Concerns and complaints

The University is committed to providing an environment in which any concerns and complaints will be treated seriously, impartially and resolved as quickly as possible. We are also committed to ensuring that a student may lodge a complaint without fear of disadvantage. In the first instance, students are encouraged (where appropriate) to attempt to resolve their concerns at a local level, by contacting their Unit Coordinator and if the Unit Coordinator can not resolve the issue then the Head of School. If you have a concern, information about who to contact for assistance is available on the Complaints and Concerns – Safe and Fair Community Unit page.

Learning support

For details of these additional services such as workshops, individual consultation for learning advice, and peer assisted learning opportunities, please visit the Study Support page.

Further information and assistance

If you are experiencing difficulties with your studies or assignments, have personal or life-planning issues, disability or illness which may affect your course of study, you are advised to raise these with the unit coordinator in the first instance.

In addition to Learning Support, there is a range of University-wide support services available to you including Student Advisers, Disability Services, and more which can be found on the Study Support and Safety, Health and Wellbeing pages from the Current Students portal of the University website.

Unit schedule

WEEK	DATE BEGINNING	TOPIC/ MODULE	ACTIVITIES	RESOURCES/ READINGS/ FURTHER INFO
1	21 February	Introduction to Marine Machinery Systems	First post to discussions	Module 1
2	28 February	Pump and Pipe Systems	Composite pipes discussion	Module 2
3	7 March	Heat Exchanger Theory		Module 3
4	14 March	Heating Ventilation and Air Conditioning (Distribution)	Ballast water discussion	Module 4 Public holiday: 14 March
5	21 March	Heating Ventilation and Air Conditioning (Psychometry)		Module 5 Census Date: 22 March
6	28 March	Reliability and Maintainability		Module 6
7	4 April	Hydraulic Systems	Class test available 8th April Class test due 10th April	Module 7
8a	11 April		Easter Break starts Thursday 14th	
Easter/mid-semester break: 14 – 20 April				
8b	21 April	Fuel and Lubricating Oil Systems	Studies resume Thursday 21st	Module 8
9	25 April	Ship Services	Conference paper due 30th April	Module 9 Public holiday: 25 April
10	2 May	Virtual Conference Week (2nd – 8th May)		
11	9 May	Hotel Services		Module 10
12	16 May	Electrical Systems		Module 11
13	23 May	Engineering of Systems	Design report due 29th May	Module 12
Study Week	30 May			